

ECON 0100 | Exercise C3 | Externalities

Smoked toffee again, with the same curves and 2 Galleon tax as Exercise C2:

$$D : P_b = 20 - \frac{1}{2}Q_d \qquad S : P_s = Q_s$$

Q1 | The Effects on Residents

a) This policy had an unintended consequence. The proliferation of smoked toffee makers meant the town saw a subsequent rise in *smoke*. Wood smoke is harmless at low levels, but can cause lung issues at the higher concentrations experienced in Hogsmeade. Use a graph to discuss the unintended effect the tax had on air quality. No need to discuss welfare.

b) Public health officials have estimated the externality to be equal to 2 Galleons per unit of smoked toffee produced. Use a graph to find the socially efficient quantity of smoked toffee and deadweight loss associated with the wood smoke. How does the socially efficient quantity compare to the tax equilibrium in Q1 of Exercise C2?

Q2 | Market Equilibrium

Hogsmeade has seen a recent proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented using the following relationships.

$$D : P_d = 100 - Q_d \qquad S : P_s = 10 + Q_s$$

The proliferation of smoked toffee makers has meant the town saw a subsequent rise in *smoke*. Wood smoke can cause lung issues and carries a cost of 10 Galleons to the health of Hogsmeade residents.

Use a graph and algebra to find equilibrium price, quantity, and DWL. Be sure to show your work.

Market Equilibrium Price: _____ Market Equilibrium Quantity: _____

Socially Efficient Quantity: _____ Deadweight Loss: _____